

Total No. of Questions : 8]

SEAT No. :

PA-1236

[Total No. of Pages : 2

[5925]-258

S.E. (Computer & Design Engineering)

COMPUTER GRAPHICS

(2019 Pattern) (Semester - III) (210244)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Differentiate between Parallel projection and perspective projection. [4]
b) What is transformation and write transformation matrix for : [4]
i) 2-D reflection with respect to line $Y = X$
ii) 3-D rotation about Y-axis
c) Perform 45° rotation of a triangle A(0, 0), B(1, 1) and C(5, 2). Find transformed coordinates after rotation, (i) About origin, (ii) About P (-1, 1). [10]

OR

- Q2)** a) What are the types of projection and write in brief about each type of projection. [4]
b) Derive 3D transformation matrix for rotation about a principal axis. [4]
c) A triangle is defined by $\begin{bmatrix} 2 & 4 & 4 \\ 2 & 2 & 4 \end{bmatrix}$. Find transformed coordinates after the following transformation. [10]
i) 90° rotation about the origin.
ii) Reflection about line $X = Y$

- Q3)** a) Explain backface detection and removal. [6]
b) Explain and compare point source and diffuse illumination. [5]
c) Compare Gouraud shading and phong shading. [6]

P.T.O.

OR

- Q4)** a) Write short note on Warnock's Algorithm [6]
b) Explain Halftone shading. [5]
c) Explain the following terms with examples: [6]
i) Color gamut
ii) Specular Reflection
iii) Diffuse reflection

- Q5)** a) Write a short note on interpolation and approximation. [4]
b) Explain blending function for B-spline curve. [7]
c) What are fractals? Explain Triadic Koch in detail. [7]

OR

- Q6)** a) Explain the Bezier curve. List its properties. [4]
b) Explain Hilbert's curve with an example. [7]
c) With suitable example write short note on the fractal line. [7]

- Q7)** a) Explain deletion of segment with suitable example. [7]
b) Define Morphing and write the applications of Morphing. [3]
c) Explain architecture of i860 [7]

OR

- Q8)** a) Write a short note on motion specification methods based on : [7]
i) Geometric and kinematics information.
ii) Specification methods based on physical information.
b) Write any three important features of NVIDIA gaming platform. [3]
c) Explain renaming of a segment with suitable example. [7]

